RESULT: generic ECDLP cost is the same on every level of a tall volcano, within ±5%

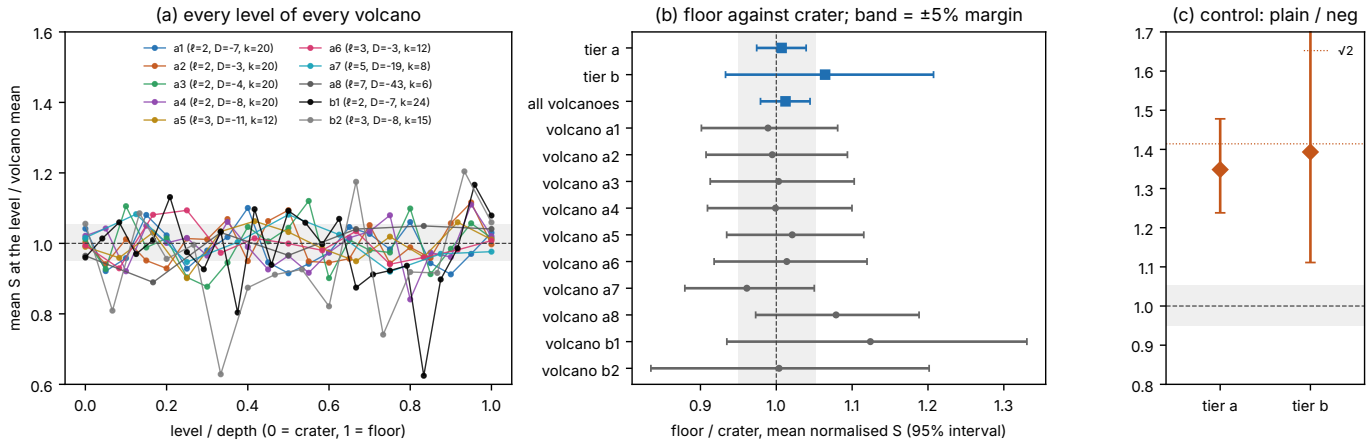
This is the pre-registered study of PROTOCOL.md. Every number below comes from TABLES.md / tables.json, written by examples/volcano_tall_analyze.rs. The figure is fig_levels.svg; the descriptive run report is reports/run-20261019/report.pdf.

Answer

Equivalent within ±5%. In tier a pooled, and in all ten volcanoes pooled, every pre-registered comparison lies inside the ±5% margin:

- floor/crater, middle/crater and floor/middle ratios;
- the crater⟶floor slope.

The distribution test (H2) finds no difference. The positive control resolves the known plain/negation gap. On these volcanoes, which way rho walks the level structure does not change its cost.



scope	solves (crater / middle / floor)	comparison	estimate	95% interval	reading
tier a (8 volcanoes)	2400 / 8804 / 2432	floor / crater	1.0070	[0.9741, 1.0394]	equivalent within the margin
tier a		middle / crater	0.9898	[0.9654, 1.0139]	equivalent within the margin
tier a		floor / middle	1.0174	[0.9881, 1.0467]	equivalent within the margin
tier a		crater⟶floor slope	+0.87%	[−2.03%, +3.75%]	equivalent within the margin
tier a		KS crater vs floor	D = 0.0199	p = 0.715	H2 not rejected
all 10 volcanoes	2625 / 9434 / 2672	floor / crater	1.0122	[0.9792, 1.0445]	equivalent within the margin
all 10		middle / crater	0.9884	[0.9647, 1.0128]	equivalent within the margin
all 10		floor / middle	1.0241	[0.9936, 1.0522]	equivalent within the margin
all 10		crater⟶floor slope	+1.16%	[−1.71%, +3.92%]	equivalent within the margin
all 10		KS crater vs floor	D = 0.0217	p = 0.542	H2 not rejected
tier b (2 volcanoes, r ≈ 2^50–2^51)	225 / 630 / 240	floor / crater	1.0644	[0.9331, 1.2073]	no difference detected; wider than the margin (scale check, as planned)

Positive control (gate passed). Σ S(rho.plain) / Σ S(rho.negation) on the same workloads and seeds:

- tier a: 1.3485 [1.2380, 1.4780] (512 pairs);
- tier b: 1.3934 [1.1111, 1.7670] (80 pairs).

Both lie above the margin. The theoretical factor is $\sqrt{2} \approx 1.414$, less the measured overhead of negation rho.

So the design resolves a 35% difference with room to spare, and bounds any level effect below about 4%.

Per volcano. None of the 40 per-volcano intervals (10 volcanoes × 4 rows) excludes the null value. At 95% about two would be expected to by chance. The widest swings are:

- a8 (ℓ=7, depth 6): floor/crater 1.079 [0.973, 1.188], slope +9.2% [−0.6, +18.4];
- a7 (ℓ=5, prime #E): floor/crater 0.961 [0.879, 1.050], slope −5.2%.

They have opposite signs, and pooling removes them.

Size. Mean raw S of rho.negation, against the floor $\sqrt{(\pi/4)} = 0.8862$:

- tier a: 0.947 [0.936, 0.958];
- tier b: 0.936 [0.902, 0.974].

The same constant holds from r ≈ 2^42 to 2^51, about 7% above the floor: negation rho's overhead, not a level effect.

What was covered

- **Volcanoes:** ten, with $\ell \in \{2, 3, 5, 7\}$ and depths 6 to 24. The tallest floors hold 8.4 and 9.6 million curves.
- **Crater types:** split, inert and ramified, including the j = 0 and j = 1728 craters.
- **Group structure:** in one volcano (a4, m = 6) E(F_p) changes from $\mathbb{Z}/64 \times \ldots$ at the crater to cyclic at the floor.
- **Sampled curves:** 608, each checked to sit at its level (class numbers within the BFS range, degree rule, blind descent) and to have the predicted group structure.
- **Solves:** 14,731 rho.negation solves at the levels, plus 592 control pairs, in 53 sessions.

Deviations, stated plainly

- **Three tier-b sessions were cut short.** A worker restart killed b1__spec_t3 (401 of 504 records), b2__spec_t3 (169 of 312) and b2__spec_t60 (105 of 120; this is b2's crater).
  - They were not re-run, as the protocol requires. Their measured records are all used.
  - Their audits report exactly two problems each: the session never reached complete, and records are fewer than planned. Every record verified and 8 of 8 replays reproduced.
  - The other 50 sessions audit OK.
  - Tier b's planned scale check is weaker than designed. Tier a and the decision rules are unaffected.
- **Analysis code edits after the first session started.** The analysis program was dry-run on the first partial records to test the code.
  - Its results were not used.

  - Three crash fixes followed: skip non-directory entries in specs/, tolerate a missing group while data was partial, and sort with NaN present.
  - No method, unit, normalisation, margin or test changed.
- **Session scheduling.** Sessions started in alphabetical rather than longest-first order (PROTOCOL.md freeze record). This affects scheduling only.

What this does and does not say

- It says the **generic** solver's cost is level-blind to within a few percent. That is what the generic-group model predicts, since all curves in the volcano share the group order r. It is now measured, at high resolution, on real tall volcanoes.
  - The earlier ±30% intervals (research/volcano_prime_layers_20261018) were too wide to say this.
  - So was the "±8%" this work once quoted, which was retracted.
- It says nothing about attacks that **use** the volcano: walking explicit isogenies to a weaker curve, endomorphism-ring–dependent index calculus, or pairings.
  - Within one volcano every curve has the same #E, so a transfer gains nothing for generic attacks: this result confirms the cost is identical at the destination.
  - Whether a non-generic attack depends on the level is a different experiment.
- The extra automorphisms of the j = 0 / j = 1728 craters are not used by ecbench's prime solvers. An automorphism-folded solver would gain $\sqrt{3}$ or $\sqrt{2}$ on those craters only. That is known theory and is not measured here.
- No claim about deployed curves. Curves are unregistered (registered=false) and are named by volcano id and level.
- Counts only (isolation L0); no wall time.